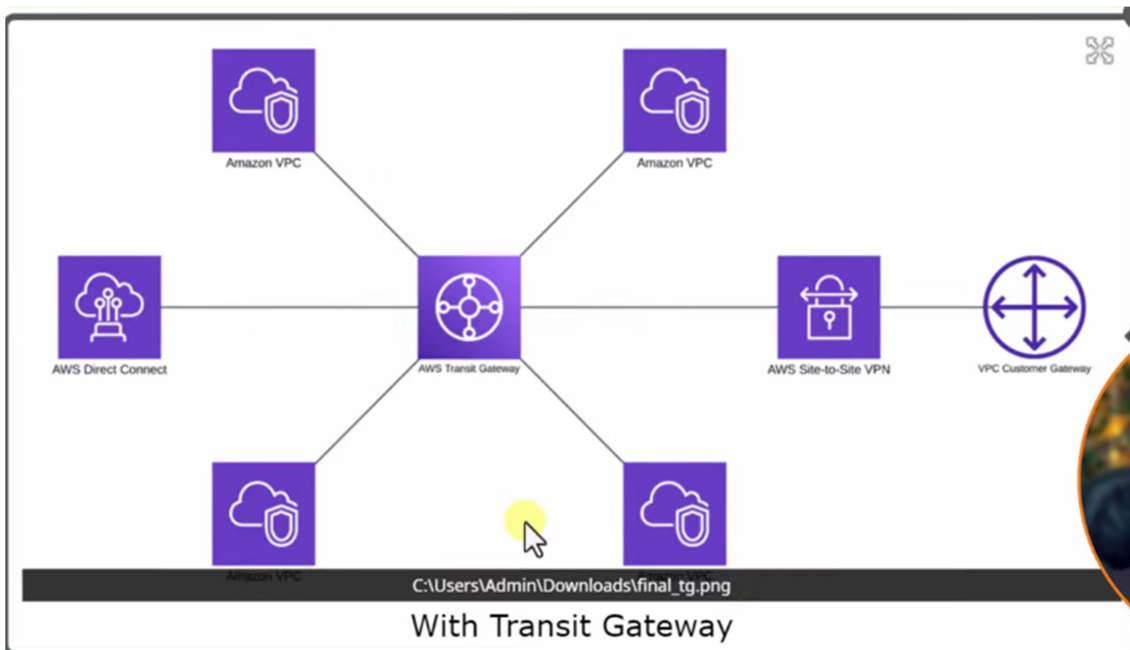
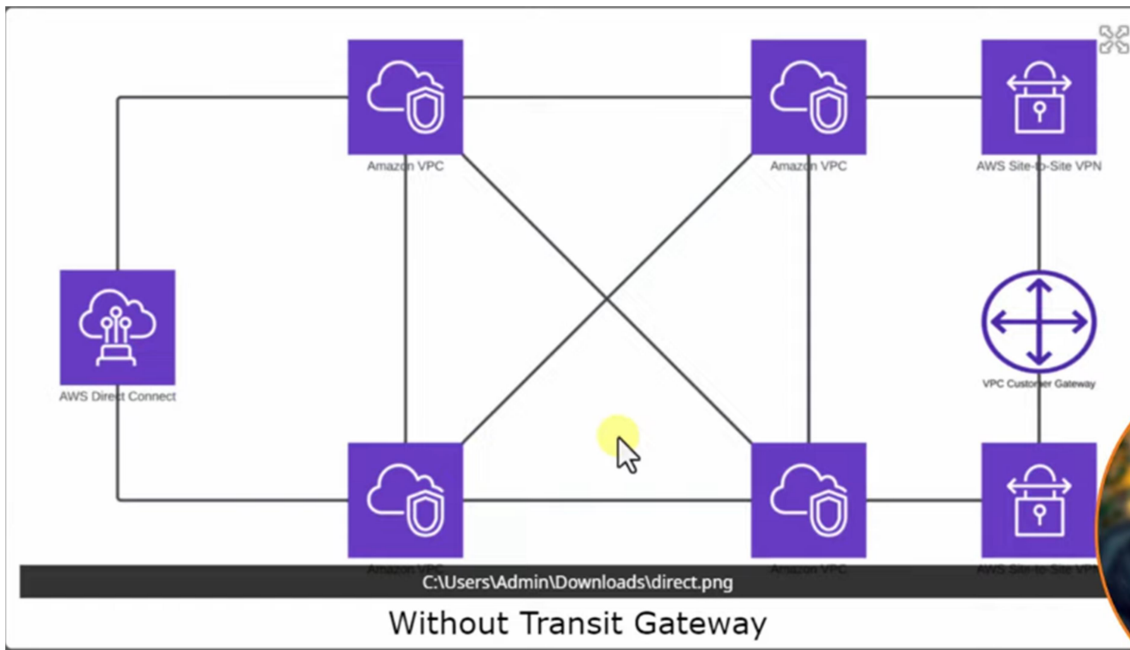
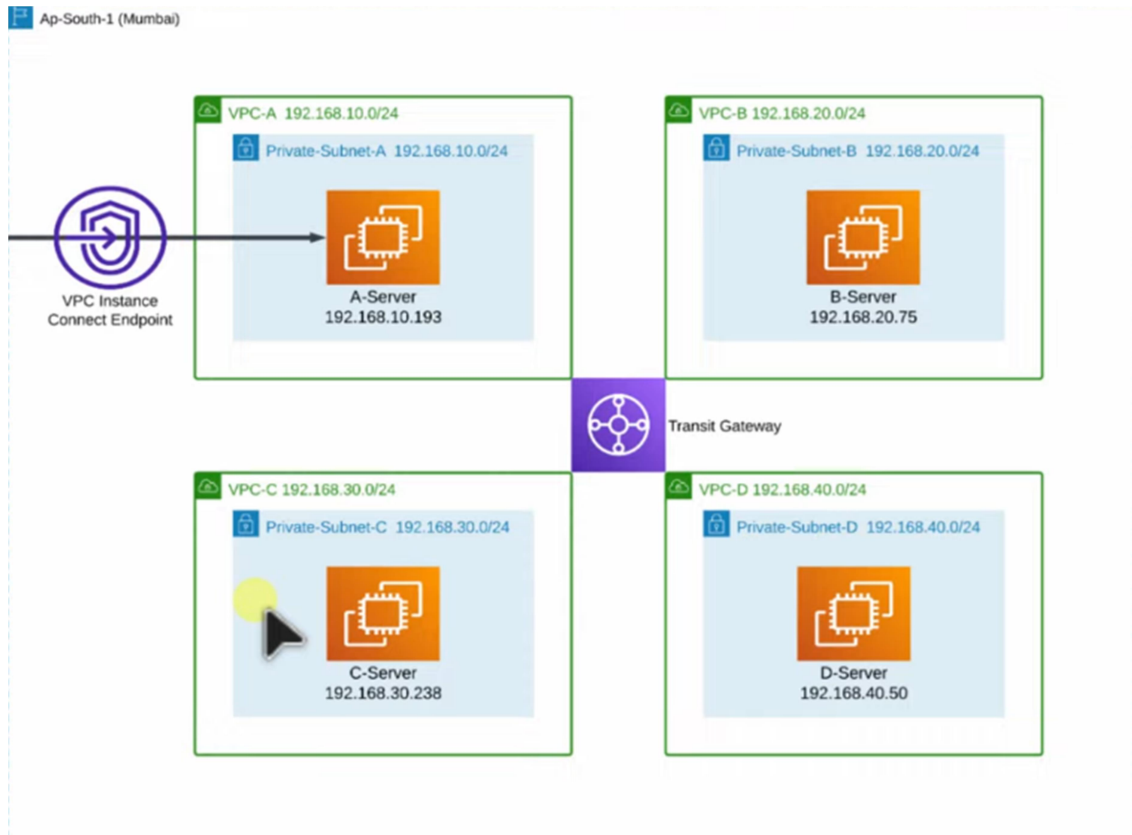






- 1- Create 4 different vpc's as per digram
- 2- Create 4 different private subnets
- 3- Create private Amazon-linux ec2 instance with no public ipv4 address with Security Group all traffic allow

Instances (4) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running X Clear filters

☐	Name	Instance ID	Instance state	Instance type	Status check
☐	C-Server	i-054b1d9a3f6d61020	Running	t2.micro	2/2 checks passed
☐	B-Server	i-069a5eca0995138ba	Running	t2.micro	2/2 checks passed
☒	A-Server	i-0654cc800b49912e4	Running	t2.micro	2/2 checks passed
☐	D-Server	i-033f056cb56930f00	Running	t2.micro	2/2 checks passed

Create endpoint : go to vpc dashboard and click endpoint

Endpoints (1) Info

Search

Name	VPC endpoint ID	VPC ID	Service name	Endpoint type
A-Server-EC2-Instance-...	eice-0a31f31e62cad440f	ypc-06a8a12825e64ab49 VPC-A	eice-0a31f31e62cad440f.bf5df978.ec2-instance-conne...	EC2 Instance Cc

Endpoint settings

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

my-endpoint-01

Service category
Select the service category

☐ AWS services
Services provided by Amazon

☐ PrivateLink Ready partner services
Services with an AWS Service Ready designation

☐ AWS Marketplace services
Services that you've purchased through AWS Mar

☒ **EC2 Instance Connect Endpoint**
An elastic network interface that allow you to connect to resources in a private subnet

☐ Other endpoint services
Find services shared with you by service name

VPC
Select the VPC in which to create the endpoint

VPC
The VPC in which to create your endpoint

ypc-06a8a12825e64ab49 (VPC-A)

► Additional settings

1:Select ec2 connect endpoint , 2: select created vpc's eg. Vpc-A

Security groups (1/2) Info

Search

Group ID	Group name	VPC ID	Description
☒ sg-06b1b8c5a89711296	default	ypc-06a8a12825e64ab49	default VPC security group
☐ sg-0f5450d75b3c42bd9	launch-wizard-1	ypc-06a8a12825e64ab49	launch-wizard-1 created 2023-11

sg-06b1b8c5a89711296 X

Subnet
Select the Subnet in which to create the endpoint

Subnet
Select the subnets in which to create the endpoint.

subnet-069221cf85bc73f77 (Private-Subnet-A)

Select security group created and subnet eg. private-subnet-A and click create endpoint

And Access instance A . go to instance and select A-server

Instances (1/4) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running X Clear filters

	Name	Instance ID	Instance state	Instance type	Status check
☐	C-Server	i-054b1d9a3f6d61020	Running	t2.micro	2/2 checks passed
☐	B-Server	i-069a5eca0995138ba	Running	t2.micro	2/2 checks passed
☒	A-Server	i-0654cc800b49912e4	Running	t2.micro	2/2 checks passed
☐	D-Server	i-033f056cb56930f00	Running	t2.micro	2/2 checks passed

Instance ID
i-0654cc800b49912e4 (A-Server)

Connection Type

☐ Connect using EC2 Instance Connect
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.

☒ **Connect using EC2 Instance Connect Endpoint**
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Private IP address
192.168.10.193

User name
Enter the user name defined in the AMI used to launch the instance. If you didn't define a custom user name, use the default user name, ec2-user.
ec2-user

Max tunnel duration (seconds)
The maximum allowed duration of the SSH connection. Must comply with the maxTunnelDuration condition (if specified) in the IAM policy.
3600
Min 1 second. Max 3600 seconds (1 hour).

EC2 Instance Connect Endpoint
Only endpoints that have completed the creation process can be selected.
eice-0a31f31e62cad440f X

Note: In most cases the default user name, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

Cancel **Connect**

Select connect using ec2 instance connect endpoint and select created endpoint id and click connect

Now connect multiple vpc's using Transit Gateway.

Go to vpc dashboard and Transit Gateway and click create transit gateway

Give name TG-cloudfolks

Configure the transit gateway

Amazon side Autonomous System Number (ASN) [Info](#)

☒ DNS support [Info](#)

☐ Security Group Referencing support [Info](#)

☒ VPN ECMP support [Info](#)

☒ Default route table association [Info](#)

☒ Default route table propagation [Info](#)

☐ Multicast support [Info](#)

Configure cross-account sharing options

☐ Auto accept shared attachments [Info](#)

Transit gateway CIDR blocks

CIDR - optional [Info](#)

leave as it is.

Now click create transit gateway

Transit gateways (1) [info](#)

☐	Name ↗	Transit gateway ID
☐	TG-CloudFolks	tgw-0e5d0bfa169e22d77

▼ Transit gateways

- Transit gateways
- Transit gateway attachments
- Transit gateway policy tables
- Transit gateway route

click transit gateway attchments. And click create transit gateway

attachment

Create transit gateway attachment [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - *optional*

Creates a tag with the key set to Name and the value set to the specified string.

A-TG

Transit gateway ID [Info](#)

tgw-0e5d0bfa169e22d77

Attachment type [Info](#)

VPC

Q

VPC

VPN

Peering Connection

Connect

☐ Security Group Referencing support [Info](#)

☐ IPv6 support [Info](#)

☐ Appliance Mode support [Info](#)

VPC ID

Select the VPC to attach to the transit gateway.

VPC attachment

Select and configure your VPC attachment.

☒ DNS support [Info](#)

☐ Security Group Referencing support [Info](#)

☐ IPv6 support [Info](#)

☐ Appliance Mode support [Info](#)

VPC ID

Select the VPC to attach to the transit gateway.

Select a VPC

- vpc-09309146a56125dbb (VPC-B)
192.168.20.0/24
- vpc-027b30d1b2d6ae5a5 (VPC-D)
192.168.40.0/24
- vpc-0ead3dade7d8a32fb (VPC-C)
192.168.30.0/24
- vpc-06a8a12825e64ab49 (VPC-A)
192.168.10.0/24

You can add up to 49 more tags.

Subnet IDs [Info](#)

Select the subnets in which to create the transit gateway VPC attachment.

☒ us-east-1a

☐ us-east-1b No subnet available

☐ us-east-1c No subnet available

☐ us-east-1d No subnet available

now click

create transit gateway attachment button

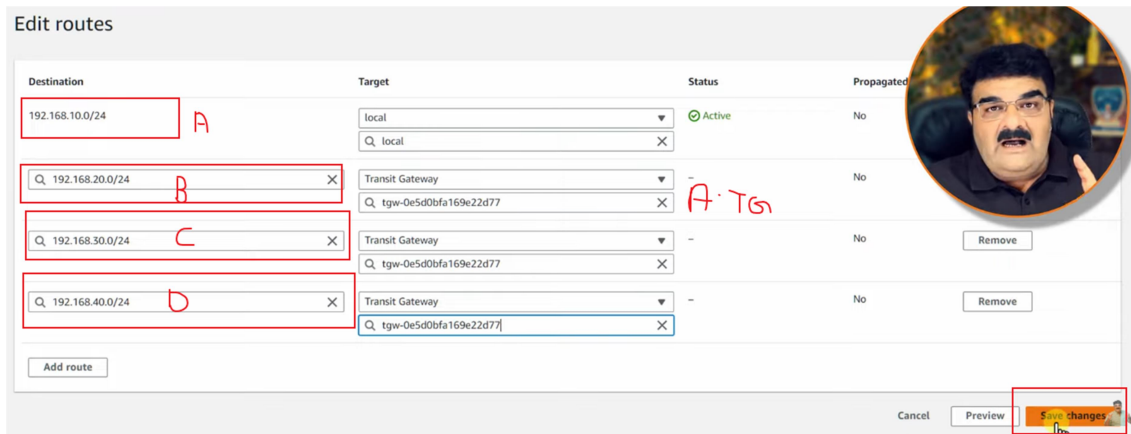
Now same for vpc B : 1-name: B-TG. 2->tg id(TG Cloudfolks) . 3>Attachment type : vpc

4> vpc id : (vpcB) . 5> subnet id: selected by default 1 (As we've created only one subnet) then click create T.G

Same for vpc-c and vpc-D. Note transit id and vpc id of that vpc's

Transit gateway attachments (4) Info							Actions
Find transit gateway attachments by attribute or tag							
☐	Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type		
☐	A-TG	tgw-attach-023347ca6193058fc	tgw-0e5d0bfa169e22d77	Available	VPC		
☐	B-TG	tgw-attach-0896667b2e2fcdde	tgw-0e5d0bfa169e22d77	Available	VPC		
☐	C-TG	tgw-attach-0a4d303a1b513dae1	tgw-0e5d0bfa169e22d77	Available	VPC		
☐	D-TG	tgw-attach-0f4a0588304749a4d	tgw-0e5d0bfa169e22d77	Available	VPC		

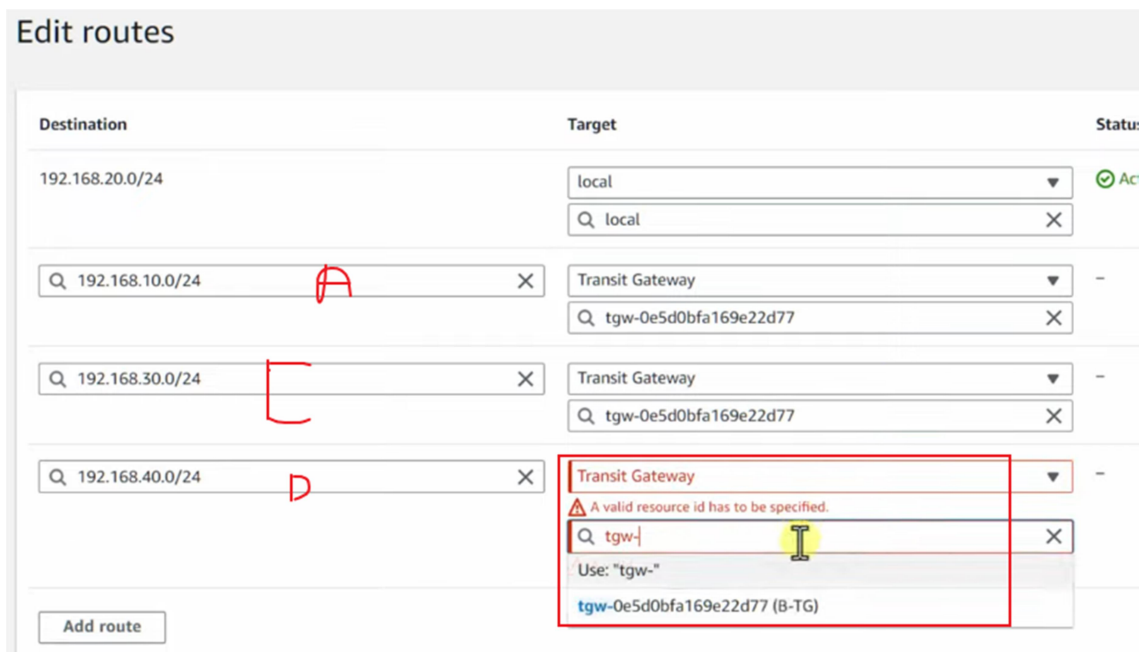
Rout Table Entry: go to vpc's route table. Select default vpc-a route table > go to route > edit route



Destination	Target	Status	Propagated
192.168.10.0/24 A	local	Active	No
192.168.20.0/24 B	Transit Gateway	-	No
192.168.30.0/24 C	Transit Gateway	-	No
192.168.40.0/24 D	Transit Gateway	-	No

Buttons: Add route, Cancel, Preview, **Save changes**

Now Select vpc -B route table > route > Edit route



Destination	Target	Status
192.168.20.0/24	local	Active
192.168.10.0/24 A	Transit Gateway	-
192.168.30.0/24 B	Transit Gateway	-
192.168.40.0/24 C	Transit Gateway	-

Buttons: Add route

Search results for 'tgw-':
 Use: "tgw-"
 tgw-0e5d0bfa169e22d77 (B-TG)

now select vpc-c and repeat process as given above diagram select id for a,b,d , and transit gateway >C-TG. and vpc-D repeat and for a,b,c and transit gateway >D-TG

Go to server-a console and ping 192.168.20.0/24 (select server-b private ip) get communicated with b and for c and d server private ip also and check the connection